

Problem 4.1 (Theoretical + MATLAB)
Bartlett test for equal covariance matrices:
derivation and numerical example

Theoretical part (can optionally be skipped)
Partial derivation of Bartlett test

Consider a multivariate normal distribution

$$X \sim N_p(\mu, \Sigma)$$

and the iid observations $x_1, x_2, \dots, x_n$.

The maximum of the likelihood function – using maximum likelihood estimators – is (see textbook)

$$\max_{\mu, \Sigma} L(\mu, \Sigma) = L(\hat{\mu}_{ML}, \hat{\Sigma}_{ML}) = \frac{1}{(2\pi)^{np/2}} e^{-np/2} \frac{1}{|\hat{\Sigma}_{ML}|^{n/2}}$$

Substituting the maximum likelihood estimator $\hat{\Sigma}_{ML}$ with the unbiased estimator

$$\hat{\Sigma} = S = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})^T$$

and correspondingly substituting n with $n - 1$ this becomes

$$L(\hat{\mu}_{ML}, S) = \frac{1}{(2\pi)^{(n-1)p/2}} e^{-(n-1)p/2} \frac{1}{|S|^{(n-1)/2}}$$

Now, this result will be used for the two-sample multivariate normal distributions case,

$$\begin{aligned}
 X_{1i} &\sim N_p(\mu_1, \Sigma_1), \quad i = 1, 2, \dots, n_1 \\
 X_{2i} &\sim N_p(\mu_2, \Sigma_2), \quad i = 1, 2, \dots, n_2
 \end{aligned}$$

to find a Likelihood Ratio test for equal covariance matrices, that is, for the hypothesis

$$H_0: \Sigma_1 = \Sigma_2 \quad \text{against} \quad H_1: \Sigma_1 \neq \Sigma_2$$

Therefore, using the result above, find an expression for the likelihood ratio

$$\Lambda = \frac{L(\hat{\mu}_1, \hat{\mu}_2, S_{\text{pooled}})}{L(\hat{\mu}_1, S_1) L(\hat{\mu}_2, S_2)}$$

where the nominator corresponds to the restricted case (H_0 true and the covariance matrix estimates can be pooled), and the denominator corresponds to the general case (H_0 not (necessarily) true and the covariance matrix estimates are not pooled).

Show that this leads to an expression of the form

$$-2 \log \Lambda = (n_1 + n_2 - 2) \log |S_{\text{pooled}}| - (n_1 - 1) \log |S_1| - (n_2 - 1) \log |S_2|$$

This can then serve as a partial explanation of the form of the Bartlett test-statistic

$$T = C \cdot [(n_1 + n_2 - 2) \log |S_{\text{pooled}}| - (n_1 - 1) \log |S_1| - (n_2 - 1) \log |S_2|] \sim X_{p(p+1)/2}^2$$

where C is the finite-population correction factor (not explained here),

$$C = 1 - \frac{2p^2 + 3p - 1}{6(p+1)} \left(\frac{1}{n_1 - 1} + \frac{1}{n_2 - 1} - \frac{1}{n_1 + n_2 - 2} \right)$$

and thus finally giving the Bartlett test of $H_0: \Sigma_1 = \Sigma_2$: Reject H_0 for $T > \frac{X_{p(p+1)/2}^2(\alpha)}{2}$

MATLAB part Numerical example

The file '**dataset_problem_4_1.mat**' contains two 24x3 data matrices containing size measurements for two samples of (non-paired) observations of 24 female and 24 male individuals of a particular species of turtles. All measurements are in unit [mm] and the columns correspond to length, width and height measurements, respectively.

It is desired to test if the covariance matrices for female and male turtles can be said to be equal, that is to test

$$H_0: \Sigma_{\text{female}} = \Sigma_{\text{male}} \text{ against } H_1: \Sigma_{\text{female}} \neq \Sigma_{\text{male}}$$

Assuming therefore that

$$X_{\text{female},i} \sim N_3(\mu_{\text{female}}, \Sigma_{\text{female}}), \quad i = 1, 2, \dots, 24$$

$$X_{\text{male},i} \sim N_3(\mu_{\text{male}}, \Sigma_{\text{male}}), \quad i = 1, 2, \dots, 24$$

with all parameters unknown

- Load the data matrices
- Calculate the descriptive statistics
- Perform a Bartlett test of H_0 as explained and partly derived above. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- perform a model check for multivariate normality including scatter matrices and QQ-plots for the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$

Problem 4.2 (MATLAB)

Two-sample comparison of multivariate mean vectors

The file '**dataset_problem_4_2.mat**' contains two 45x2 data matrices containing size measurements for two samples of (non-paired) observations of 45 female and 45 male individuals of a particular species of birds, namely hook-billed kites. All measurements are in unit [mm] and the columns correspond to tail length and wing length, respectively.

It is desired to test for equality of the mean vectors for the populations of male and female hook-billed kites.

Preliminary examination of samples, part 1: Bartlett test

First it shall be established whether the sample covariance matrices for the two samples can be pooled, that is to test

$$H_{0,\Sigma}: \Sigma_{\text{female}} = \Sigma_{\text{male}} \text{ against } H_1: \Sigma_{\text{female}} \neq \Sigma_{\text{male}}$$

Assuming therefore that

$$X_{\text{female},i} \sim N_2(\mu_{\text{female}}, \Sigma_{\text{female}}), \quad i = 1, 2, \dots, 45$$
$$X_{\text{male},i} \sim N_2(\mu_{\text{male}}, \Sigma_{\text{male}}), \quad i = 1, 2, \dots, 45$$

with all parameters unknown

- Load the data matrices
- Calculate the descriptive statistics
- Perform a Bartlett test of $H_{0,\Sigma}$ as explained in previous problem. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- Conclude

Preliminary examination of samples, part 2: check of multivariate normal model

Perform a model check for multivariate normality:

- Make plots of the multivariate scatter matrices plus marginal histograms and boxplots
- Calculate the squared sample Mahalanobis distances, $d_i^2 = (x_i - \bar{x})^T S^{-1} (x_i - \bar{x})$, and make QQ-plots versus the relevant χ^2 distribution for check of approximate linearity for both female and male samples
- Conclude

Test for equality of the mean vectors for female and male populations

Thus having (hopefully) reasonably well established that

$$\begin{aligned} X_{\text{female},i} &\sim N_2(\mu_{\text{female}}, \Sigma_{\text{female}}), \quad i = 1, 2, \dots, 45 \\ X_{\text{male},i} &\sim N_2(\mu_{\text{male}}, \Sigma_{\text{male}}), \quad i = 1, 2, \dots, 45 \end{aligned}$$

with

$$\Sigma_{\text{female}} \cong \Sigma_{\text{male}},$$

the main purpose of this investigation can be addressed, namely to test for equality of mean vectors for the male and female populations of hook-billed kites, that is to test

$$H_{0,\mu}: \mu_{\text{female}} = \mu_{\text{male}} \text{ against } H_1: \mu_{\text{female}} \neq \mu_{\text{male}}$$

- Perform an exact test of the hypothesis $H_{0,\mu}$. Use level of significance $\alpha = 0.05$ but calculate also the p-value for the test
- Perform an large-sample approximate test of the same hypothesis and use again $\alpha = 0.05$ but calculate also the p-value
- Compare the two test-results and conclude

Confidence region and intervals for difference in mean vectors

Turning to confidence region and intervals, still using $\alpha = 0.05$,

- Calculate (and optionally plot) the $100(1 - \alpha)\%$ confidence region for the difference in mean vectors, $(\mu_{\text{female}} - \mu_{\text{male}})$. Is $[0,0]$ inside the confidence region?
- Calculate simultaneous $100(1 - \alpha)\%$ confidence intervals for $(\mu_{\text{female}} - \mu_{\text{male}})$
- Calculate marginal $100(1 - \alpha)\%$ confidence intervals for $(\mu_{\text{female}} - \mu_{\text{male}})$
- Calculate Bonferroni $100(1 - \alpha)\%$ confidence intervals for $(\mu_{\text{female}} - \mu_{\text{male}})$
- Compare the various confidence intervals and explain
- Also compare with the hypothesis test results